ARASH PARSA

1(510) 626-0266 | arash.mehrabi.parsa@gmail.com

Data scientist building ML systems for healthcare: clinical NLP, EHR modeling, and rigorous evaluation. Experience with Python, SQL/PostgreSQL, and production LLM testing.

PROFILE

- Focused, hardworking, and result oriented
- Innovative and analytical problem solver
- Adept in learning and application of new concepts and technologies
- High organizational IQ, great team player who easily connects with team members
- Natural Leadership aptitude

TECHNICAL SKILLS

- **Programming:** Python, R, SQL, C++, MATLAB, Bash, JavaScript
- **Data Science & Statistics:** statistical modeling, hypothesis testing, A/B testing, experimental design, engineering, time series analysis, cross-validation
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, PostgreSQL, MS SQL Server, Redis, Excel, dashboard development
- **Machine Learning & AI:** scikit-learn, XGBoost, TensorFlow, PyTorch, HuggingFace Transformers, LLMs, classification, regression, clustering, NLP, deep learning, CNNs, graph neural networks, predictive modeling
- **Pipeline, Cloud & Reproducibility:** Nextflow, Docker, Git, Linux, AWS, RE

DOMAIN KNOWLEDGE

- **Bioinformatics & Omics:** Scanpy, Squidpy, cell2location, SpatialDE, DESeq2, clusterProfiler, STAR, featureCounts, FastQC, Trimmomatic, KEGG, Gene Ontology, sequence alignment, genome analysis, primer design
- **Spatial Transcriptomics:** 10x Visium, spatially variable gene detection, cell-type deconvolution, spatial clustering, tissue domain identification
- **Clinical NLP & Health Informatics:** BioBERT, named entity recognition, model cascades, benchmark design, MIMIC-III/IV, n2c2, EHR data analysis, patient phenotyping, clinical decision support, health data standards
- **Molecular & Synthetic Biology:** PCR, Gibson Assembly, yeast transformation, Cre-Lox, gel electrophoresis, Sanger sequencing, CRISPR, plasmid cloning, metabolic pathway engineering, gene cassette design, heterologous expression, recombinant DNA

EDUCATION

- **MS** in Data Science/Analytics **Georgia Tech**
- **BS** in Bioengineering/Bioinformatics **UC Santa Cruz**

CERTIFICATIONS *(Forage Job Simulations)*

- **BCG X** – Data Science Job Simulation – June 15, 2026 | Enrolment: fciegf6TTYjpzDNA2
- **J.P. Morgan** – Quantitative Research Job Simulation – June 15, 2026 | Enrolment: RJeiTdDMcvgsrqKfE
- **Deloitte** – Data Analytics Job Simulation – June 15, 2026 | Enrolment: ErcGTooquAqG4iJof
- **British Airways** – Data Science Job Simulation – June 20, 2026 | Enrolment: GtJ3QQsQ4qrrr6Nzf

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WORK EXPERIENCE

Aug 2026 – Present **AI Evaluation & QA Intern**
Veru365 | Fintech

- Red-teamed production LLM financial agents; found compliance failures and verified fixes
- Designed adversarial evaluation datasets (197 refusal prompts; sycophancy and scope-evasion suites) adapted from published AI-safety benchmarks
- Built an LLM-as-judge pipeline: 97% accuracy at about 7x lower cost than a commercial testing tool
- Shipped bug fixes to the production backend of a fintech AI platform through code-reviewed pull requests
- Ran release QA and benchmarked AI testing vendors on cost and accuracy

Jan 2026 – Present **Machine Learning Research Collaborator – Point-in-Time Clinical Trial Matching**
Dr. Vivek Rudrapatna's Clinical Data Science Group – UCSF | San Francisco, CA

- Opened a second workstream in the same collaboration and reframed trial matching as a *point-in-time* decision – "from only the notes written before hour *t*, is this patient screenable now?" – rather than retrospective derivation from discharge summaries, which are unavailable at the moment a screening decision is actually made
- Designed a criterion-conditioned attention model over timestamped note streams with per-head learned recency decay: a frozen encoder embeds and caches each note chunk once, so only a 1.8×10^5-parameter head trains on CPU with no GPU and no LLM API at inference. Encoding is paid once per patient rather than once per patient×trial, so adding the 50th trial costs one criterion encode instead of another full pass over the census
- Beat both open-weight LLM generations at $\sim 10^7$ times less inference compute: AUROC **0.753** [0.687, 0.810] versus 0.634 (Qwen2.5-7B) and 0.643 (Qwen3-8B) zero-shot over 363 paired patient–criterion decisions across 312 MIMIC-III patients, paired bootstrap deltas +0.119 and +0.109 with confidence intervals excluding zero; also clears an XGBoost-on-structured-labs baseline at 0.722
- Built the evaluation to attack its own result: identified that four of five eligibility criteria are defined by the very lab value they would be predicted from, so a naive run scores AUROC 1.000 and measures nothing, then reported results only on the criterion requiring clinical judgment and added leave-one-concept-out and note-volume-only null arms to catch accuracy that comes from counting notes rather than reading them
- Extended the same cached representations to a second task graded by an answer key the model never sees – whether a criterion is *ascertainable* at *t*, labeled exactly and at zero annotation cost from lab result timestamps (0.601 ± 0.008 over 2,505 decisions versus 0.504 for the volume-only control) – and showed that 14.5% of patient–criterion pairs change ascertainability status between hour 6 and hour 24, most of it evidence *expiring* rather than resolving. Split-conformal thresholding gives a distribution-free recall guarantee; manuscript in preparation with UCSF corpus validation planned

Jan 2026 – Present **Data Science Research Assistant in Clinical Research**
In collaboration with Stanford-affiliated researchers | San Francisco, CA

- Trichoscopic Oculomics – built deep learning classification pipeline integrating spatial transcriptomics and metabolic epidemiology features to reframe androgenetic alopecia as a systemic screening opportunity
- Endometriosis Severity Prediction – developed supervised ML models (random forest, gradient boosting, logistic regression) to phenotype and preoperatively predict endometriosis severity from clinical and demographic features extracted from EHR-derived datasets
- Performed end-to-end clinical data preprocessing: missing data imputation, feature encoding, class imbalance correction (SMOTE), and cross-validated model evaluation; presented results to clinical research stakeholders

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WORK EXPERIENCE (Continued)

Dec 2025 – Mar 2026 **Machine Learning & Data Science Research Assistant**
Washington Lab, UCSF | San Francisco, CA

- Systematic review study: cataloged model architectures (CNNs, LSTMs, transformers), dataset characteristics, and performance benchmarks across 50+ studies on home video-based ML classification of autism and ADHD in pediatric populations
- Developed Python scripts to extract, standardize, and visualize study metrics (sensitivity, specificity, AUC) across heterogeneous reporting formats, enabling quantitative cross-study comparison for clinical decision support
- Manuscript to be submitted for publication in the near future

Sept 2022 – Present **Machine Learning Scientist**
Optelligence | Berkeley, CA

- Designed and trained supervised regression and classification models (XGBoost, neural networks) for performance prediction, cutting physical testing cycles by 50%
- Built end-to-end data preprocessing pipelines in Python (Pandas, NumPy, scikit-learn) including feature engineering, outlier detection & normalization across heterogeneous datasets
- Applied CNNs and graph neural networks in TensorFlow and PyTorch to extract latent structure from high-dimensional experimental measurements
- Constructed A/B testing framework with statistical significance testing (t-tests, bootstrap CIs) to evaluate model variants, improving deployment efficiency by 25%
- Wrote SQL queries to extract and join data from PostgreSQL databases; created data visualizations and dashboards to communicate findings to cross-functional stakeholders

Aug 2021 – May 2022 **R & D Engineer**
Amgen & Thermofisher | Thousand Oaks, CA & Pleasanton, CA

- Automated quality control workflows using Python and SQL, reducing manual review time and maintaining 99.8% assay accuracy across 100+ daily COVID-19 diagnostic samples
- Developed statistical process control (SPC) dashboards to monitor HPLC/UPLC instrument drift, enabling early anomaly detection and reducing reagent waste by $50K annually
- Performed multivariate data analysis on chromatography outputs using R and Excel; identified systematic variance sources, optimized parameters through iterative experimentation, and presented findings and recommendations to cross-functional clinical and laboratory stakeholders